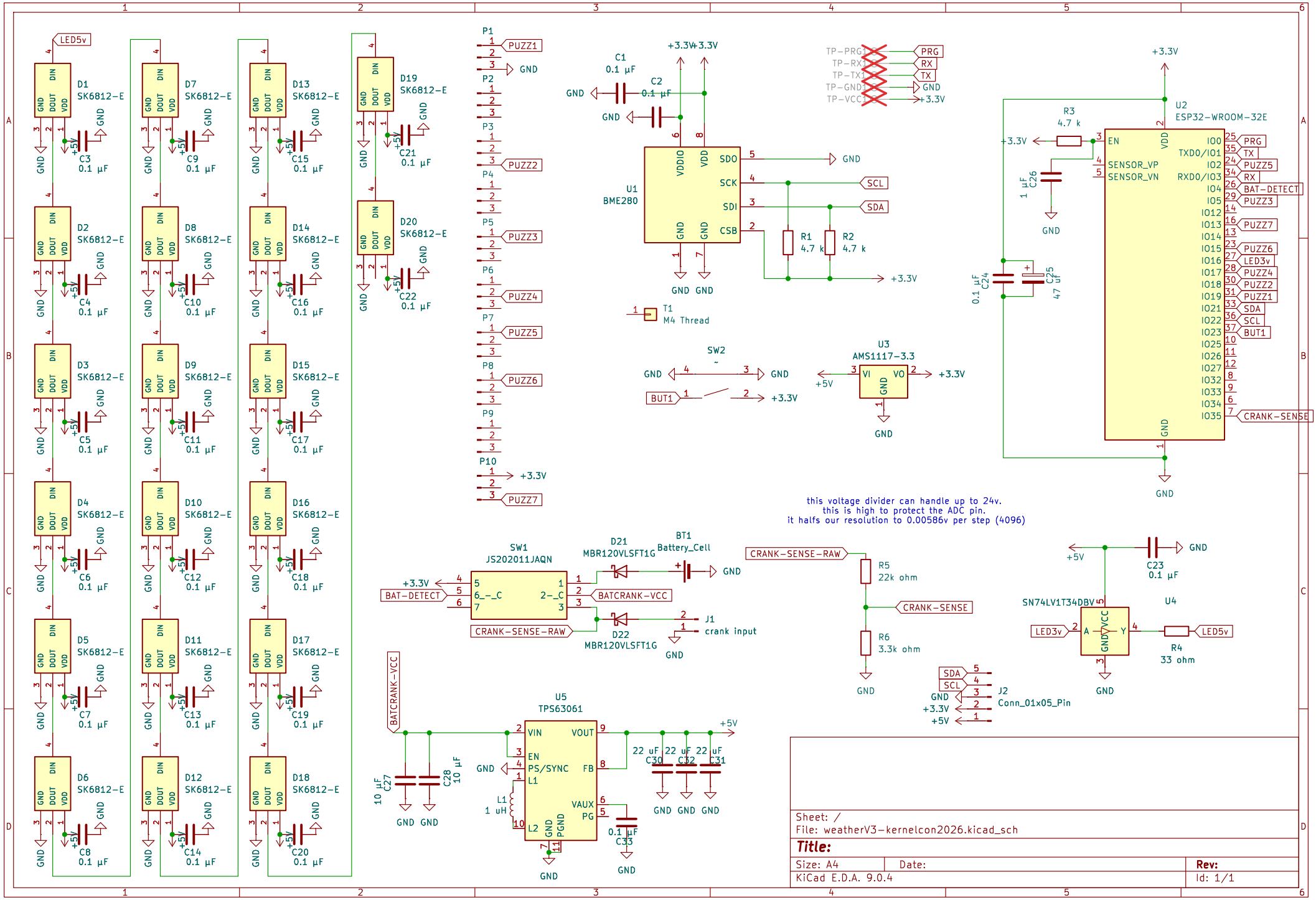

LED5v
D1 SK6812-E
D2 SK6812-E
D3 SK6812-E
D4 SK6812-E
D5 SK6812-E
D6 SK6812-E
D7 SK6812-E
D8 SK6812-E
D9 SK6812-E
D10 SK6812-E
D11 SK6812-E
D12 SK6812-E
D13 SK6812-E
D14 SK6812-E
D15 SK6812-E
D16 SK6812-E
D17 SK6812-E
D18 SK6812-E
D19 SK6812-E
D20 SK6812-E
C1 0.1 µF
C2 0.1 µF
C3 0.1 µF
C4 0.1 µF
C5 0.1 µF
C6 0.1 µF
C7 0.1 µF
C8 0.1 µF
C9 0.1 µF
C10 0.1 µF
C11 0.1 µF
C12 0.1 µF
C13 0.1 µF
C14 0.1 µF
C15 0.1 µF
C16 0.1 µF
C17 0.1 µF
C18 0.1 µF
C19 0.1 µF
C20 0.1 µF
C21 0.1 µF
C22 0.1 µF
P1 PUZZ1
P2
P3 PUZZ2
P4
P5 PUZZ3
P6 PUZZ4
P7 PUZZ5
P8 PUZZ6
P9
P10 +3.3V
PUZZ7
U1 BME280
VDDIO VDD SDO SCK SDI CSB GND GND
SCL
SDA
R1 4.7 k
R2 4.7 k
T1 M4 Thread
TP-PRG TP-RX TP-TX TP-GND TP-VCC
PRG RX TX GND +3.3V
SW2
BUT1
U3 AMS1117-3.3
VI VO GND
U2 ESP32-WROOM-32E
R3 4.7 k
C24 0.1 µF
C25 47 uF
C26 1 µF
EN SENSOR_VP SENSOR_VN VDD GND
IO0 TXD0/IO1 IO2 RXD0/IO3 IO4 IO5 IO12 IO13 IO14 IO15 IO16 IO17 IO18 IO19 IO21 IO22 IO23 IO25 IO26 IO27 IO32 IO33 IO34 IO35
PRG TX PUZZ5 RX BAT-DETECT PUZZ3 PUZZ7 PUZZ6 LED3v PUZZ4 PUZZ2 PUZZ1 SDA SCL BUT1 CRANK-SENSE
this voltage divider can handle up to 24v.
this is high to protect the ADC pin.
it halfs our resolution to 0.00586v per step (4096)
SW1 JS202011JAQN
BAT-DETECT
CRANK-SENSE-RAW
BATCRANK-VCC
D21 MBR120VLSFT1G
D22 MBR120VLSFT1G
BT1 Battery_Cell
J1 crank input
R5 22k ohm
R6 3.3k ohm
CRANK-SENSE
C23 0.1 µF
U4 SN74LV1T34DBV
LED3v
LED5v
R4 33 ohm
J2 Conn_01x05_Pin
SDA SCL GND +3.3V +5V
U5 TPS63061
VIN VOUT EN PS/SYNC FB L1 L2 VAUX PG GND PGND
L1 1 uH
C27 10 µF
C28 10 µF
C30 22 uF
C32 22 uF
C31 22 uF
C33 0.1 µF
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Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.4
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